

Microsoft Teams

Documentation

1. Project Overview

The Microsoft Teams Replica is a beginner-friendly web application that helps users communicate, collaborate, and share information online. It combines chat, team collaboration, meetings, and file sharing in a single platform.

2. Problem Statement

Many students and organizations use different applications for communication, meetings, and file sharing. Managing multiple tools can be difficult. This project provides a single platform where users can chat, conduct meetings, and collaborate easily.

3. Objectives

- Enable user communication
- Create teams and channels
- Support online meetings
- Share files and information
- Improve collaboration among users

4. User Roles

- User
- Team Member
- Team Admin
- Administrator

5. Technology Stack

Frontend: HTML, CSS, Bootstrap, JavaScript/React

Backend: Node.js, Express.js

Database: MySQL

Tools: Git, GitHub, VS Code, Postman

6. Core Modules

- Authentication Module
- Personal Chat Module
- Team Management Module
- Channel Management Module
- Meeting Module
- File Sharing Module
- Notification Module

7. Module Description

Authentication handles registration and login. Personal Chat enables one-to-one messaging. Team

Management allows users to create and join teams. Channels help organize discussions. Meetings support video communication. File Sharing allows document exchange. Notifications keep users updated.

8. Database Design

- Users
- Teams
- Channels
- Messages
- Meetings
- Files

9. System Workflow

User Registration → Login → Create/Join Team → Chat with Members → Schedule Meeting → Share Files → Receive Notifications

10. Implementation Plan

Phase 1: Authentication

Phase 2: Chat System

Phase 3: Team Management

Phase 4: Channel Management

Phase 5: Meetings

Phase 6: File Sharing

Phase 7: Testing and Deployment

11. Security Features

- User Authentication
- Encrypted Passwords
- Secure Database Connection
- Access Control

12. Future Enhancements

- Mobile Application
- Real-Time Notifications
- AI-based Meeting Summary
- Advanced Analytics

13. Expected Benefits

- Better communication
- Improved collaboration
- Easy file sharing
- Centralized platform for teamwork

14. Conclusion

The Microsoft Teams Replica helps users communicate and collaborate through chat, meetings, teams, and file sharing. The project is simple, beginner-friendly, and suitable for learning full-stack web development.